C3-Diff: Super-resolving Spatial Transcriptomics via Cross-modal Cross-content Contrastive Diffusion Modelling

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1. Abstract & Motivation

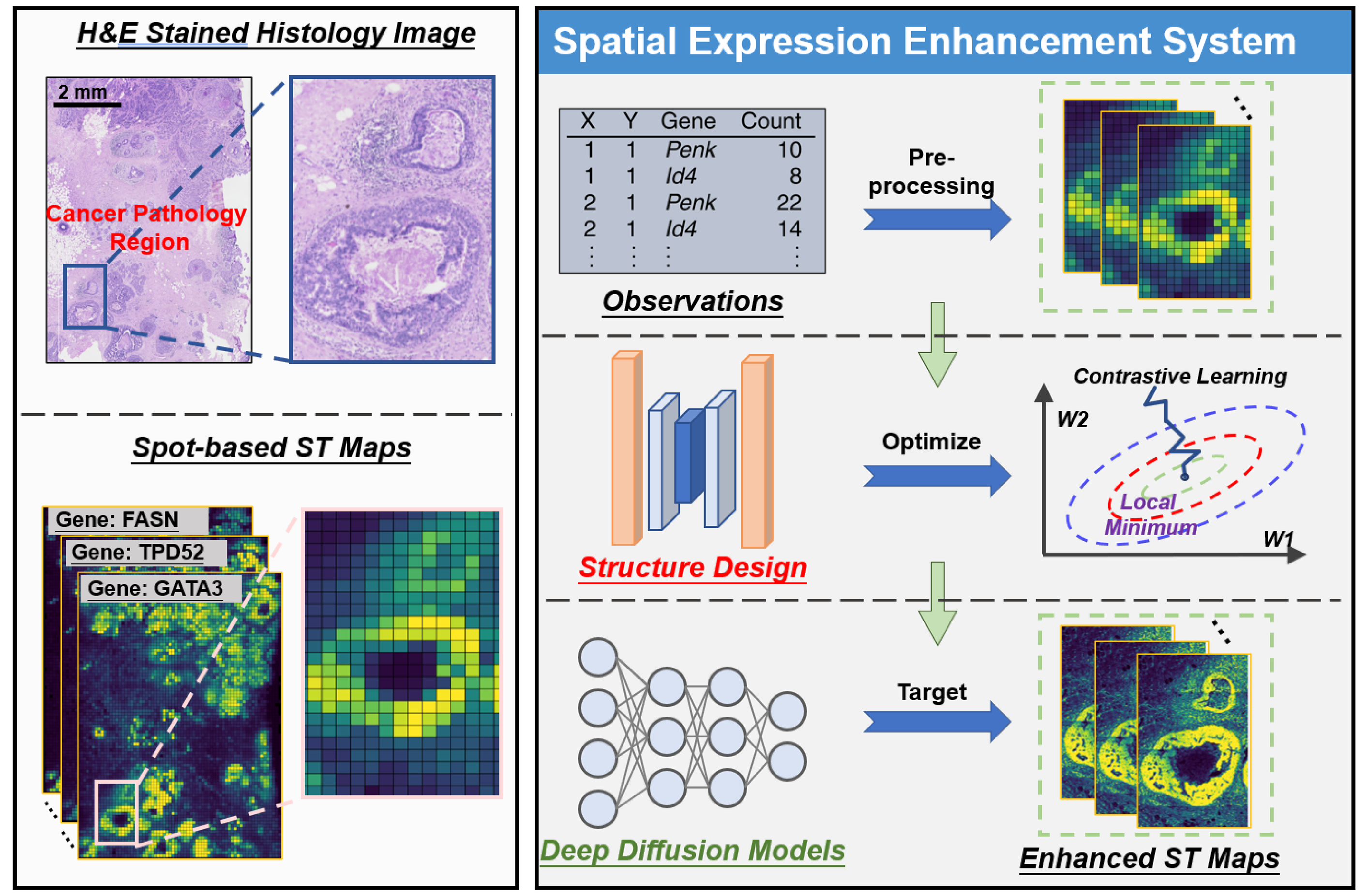
Spatial transcriptomics (ST) measures tissue context but often suffers from low spatial resolution and low sequencing sensitivity. We present C3-Diff, a cross-modal cross-content contrastive diffusion framework that enhances ST resolution through self-guided diffusion leaming. By exploiting co-located ST and high-resolution (HR) cross-modail features, the model embeds gene spots with hyperspheres, and supports incomplete-tely overlapping cross-donsain ST. C3-Diff consistently outperforms top ST super-resolution and spatial diffusion baselines across ST, cancer subtypes, and spatial omics, achieving while improving downstream tasks including single-cell expression prediction, gene expression correlation analysis, and cell-type localization.

Why this matters

- Low-resolution spots limit analyses, insights, calls and tasks.
- Hypermapping cells? Unfortunately no.
- High-resolution ST technologies remain costly and often below low capture sensitivity.
- Bridging legacy low SR assays available and provide rich insights for large-scale applications.

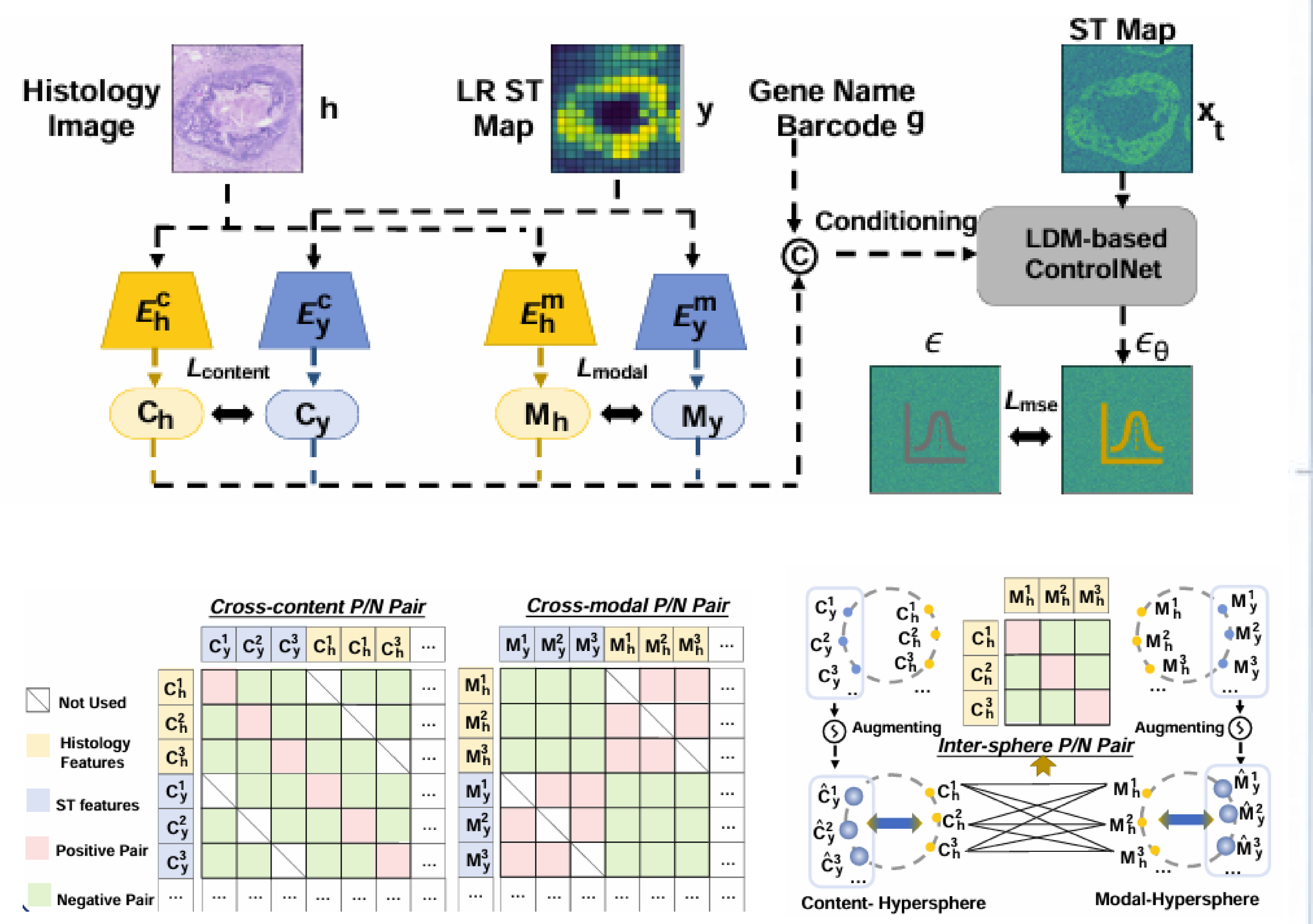
Key Contributions

- Cross-modal cross-content contrastive learning to capture both modality-specific and modality-invariant information.
- A bridging-based information augmentation on the unit hypersphere to inspire information loss in ST.
- Dynamic instance and integration learning for an intrinsic LR ST enabling practical and useful deployment.



(a) Input (b) ST enhancement system

2. Method



- **Inputs:** Histology image I_A, low-resolution ST map x, and gene barcode g
- **Backbone:** LDM-based ControlNet for conditional generation of high-resolution ST.
- **Dual contrastive losses:** explicitly encode cross-modal cross-content invariance and content-invariant representations.
- **Losses:** $\mathcal{L} = \alpha\mathcal{L}_{con} + \beta\mathcal{L}_{bri} + \gamma\mathcal{L}_{cov} + \delta\mathcal{L}_{negative}$
- **Cross-modal inpainting** ensures missing ST features using histology-based similarity during training.

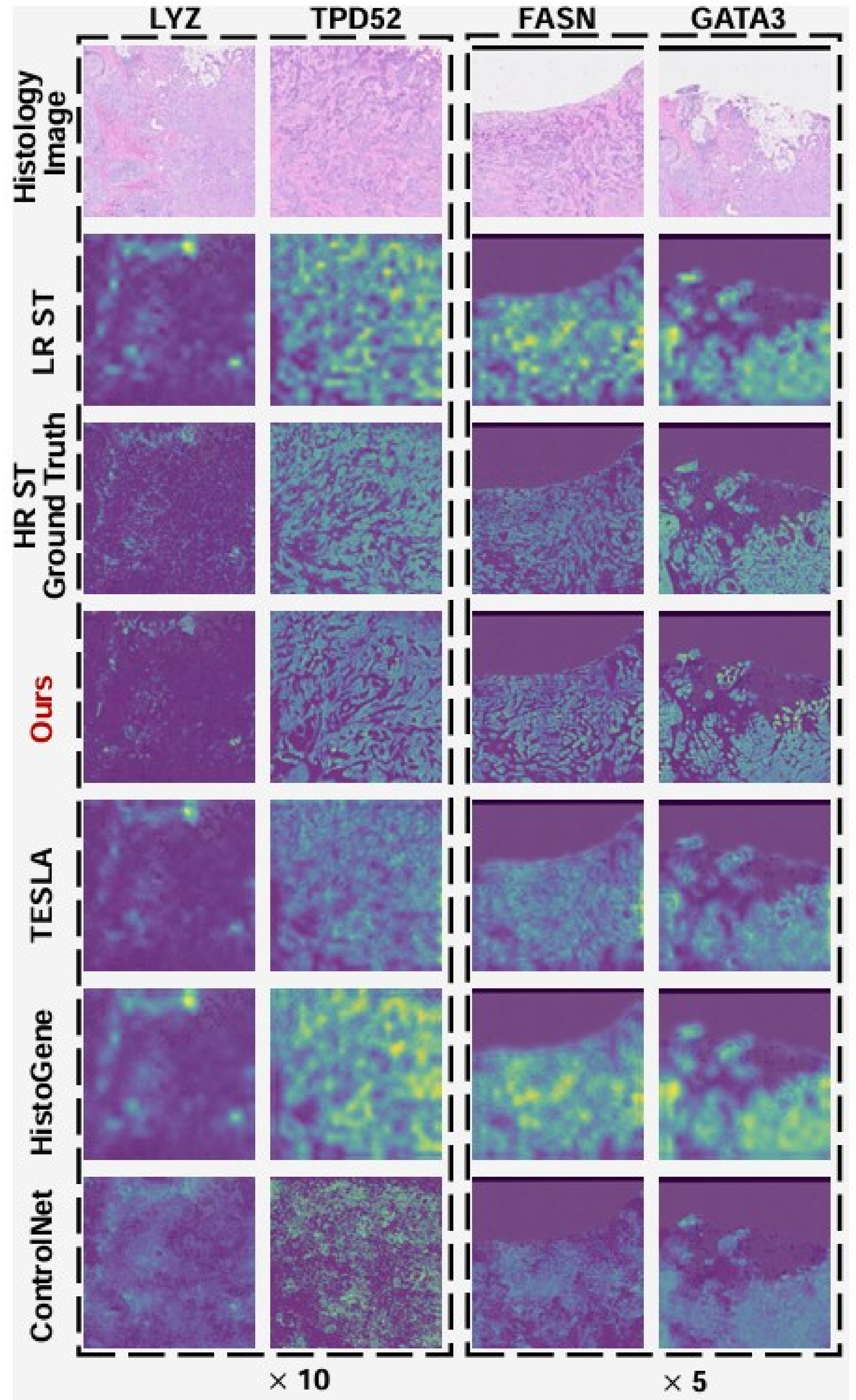
Core Idea

Bridge morphology and gene expression through contrastive diffusion modelling.

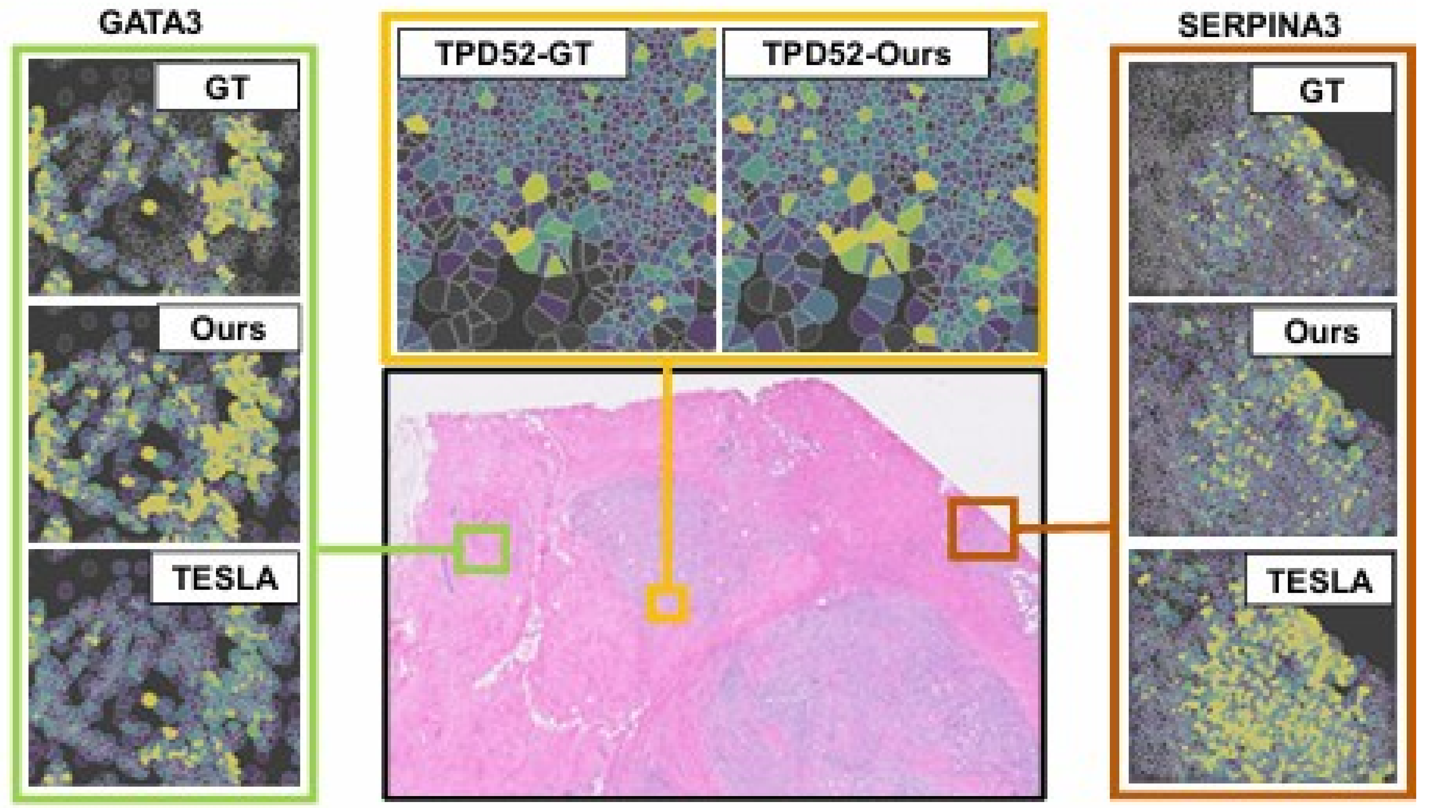
3. Experimental Results

Dataset	Attributes		Breast-Xenium				Breast-SGE			
Scale			5×		10×		5×		10×	
	For ST	IMT**	RMSE	PCC	RMSE	PCC	RMSE	PCC	RMSE	PCC
U-Net			0.385	0.178	0.407	0.192	0.356	0.484	0.455	0.545
U-Net++			0.302	0.224	0.289	0.314	0.376	0.512	0.434	0.509
AttenU-Net			0.385	0.162	0.423	0.196	0.337	0.402	0.434	0.367
LDM			0.317	0.286	0.296	0.331	0.315	0.493	0.386	0.493
ControlNet			0.219	0.315	0.248	0.324	0.286	0.547	0.324	0.509
Uni-Control		✓	0.252	0.365	0.240	0.343	0.294	0.508	0.339	0.545
HistoGene	✓		0.235	0.262	0.271	0.328	0.315	0.501	0.342	0.508
iStar	✓		0.248	0.352	0.247	0.352	0.296	0.512	0.338	0.526
TESLA	✓		0.196	0.314	0.235	0.386	0.285	0.548	0.312	0.513
BLEEP	✓		0.242	0.350	0.245	0.372	0.293	0.482	0.296	0.508
Diff-ST	✓		0.168	0.346	0.184	0.392	0.224	0.542	0.247	0.525
Ours (no LR ST)*	✓	✓	0.112	0.376	0.148	0.410	0.160	0.571	0.196	0.550
Ours	✓	✓	**0.094**	**0.386**	**0.137**	**0.432**	**0.146**	**0.582**	**0.175**	**0.575**

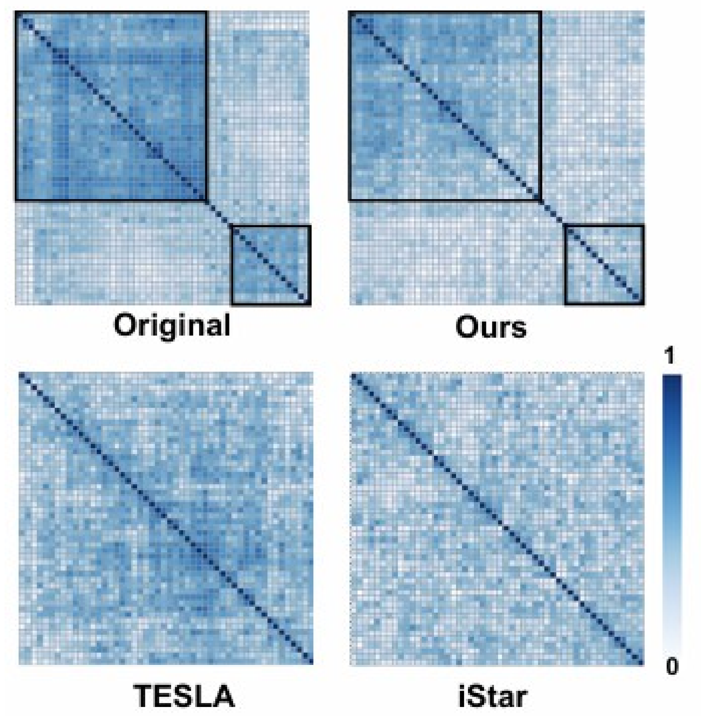
C3-Diff remains competitive even without LR ST at test time.

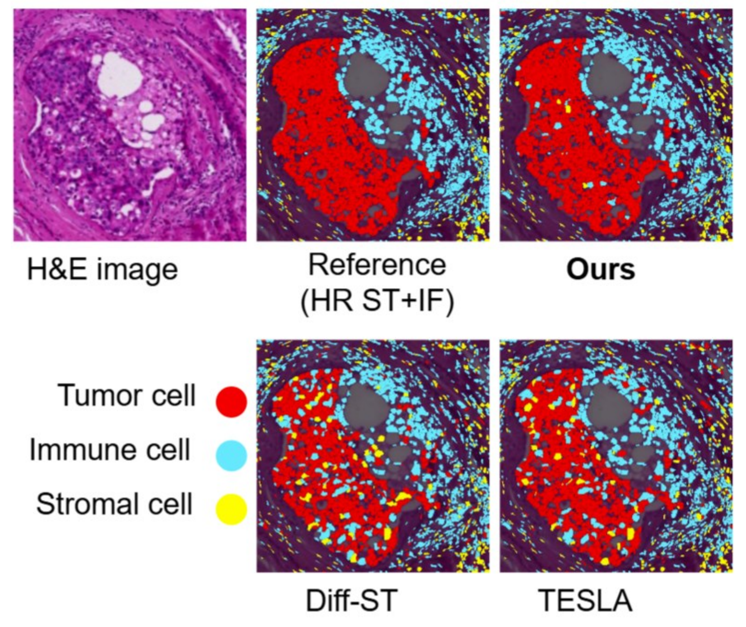


4. Downstream Validation & Impact



Gene Expression Correlations





5. Conclusions

- First cross-modal cross-content contrastive diffusion model for ST
- Consistently outperforms leading SR and ST super-resolution and cross-modal diffusion baselines.
- Preserves accurate biological analysis and supports lower-cost large-scale spatial profiling.
- Supports diverse real-world legacy spatial datasets.